

PRACTICE EXAM

Difficulty: MEDIUM

Questions: 10

Electronics Exam

Instructions: Please answer all questions to the best of your ability.

Multiple Choice Questions (4 points each)

Instructions: Choose the best answer for each question.

Question 1: According to the introduction, what is a concept that this lab is meant to reintroduce to the student?

- A) Ohm's Law
- B) Kirchhoff's Laws
- C) Thevenin's Theorem
- D) Superposition Principle

Question 2: In the circuit under test, what components are connected in parallel on the right side of the circuit, before the capacitor is introduced?

- A) Two resistors and a capacitor
- B) A resistor and a voltage source (VCC)
- C) Two capacitors
- D) A signal generator and a resistor

Question 3: According to the passage, what software was used to simulate the circuit?

- A) MATLAB
- B) PSpice
- C) Multisim
- D) LTspice

Question 4: What was the peak-to-peak voltage of the sinusoidal signal used in the simulation?

- A) 2V
- B) 4V
- C) 5V
- D) 10V

Short Answer Questions (6 points each)

Instructions: Answer each question in 2-3 complete sentences.

Question 5: Briefly describe the objective of the lab, as stated in the provided text.

Question 6: Explain what Figure 3 represents in terms of the circuit's configuration.

Question 7: According to conclusion C1, what are two possible reasons cited for discrepancies between Table P1 and Table E3?

Problem-Solving Questions (10 points each)

Instructions: Provide a detailed analysis and solution for each problem. Show all your work.

Question 8: Based on the information in the "Experimental Results" section, describe the process used to obtain the v_I voltages in Table E1.

Question 9: Explain how the principle of superposition is used to justify the waveform of V_O in E3 (Figure 6), referencing information from experiments E1 and E2. Quantify this explanation using values provided in the text.

Question 10: If the resistance of the signal generator was negligibly small instead of 50 ohms, how would you expect the values in Table E1 to change, specifically the v_I (rms) value? Explain your reasoning.